

MATHS TEST — SET 2 (NEXT LEVEL)

Class – IX (UP Board Pattern)

Chapters: Real Numbers & Algebraic Identities

Time Allowed: 1 Hour 30 Minutes

Maximum Marks: 35

General Instructions:

1. All questions are compulsory.
2. Calculator is not permitted.
3. Show necessary steps for full marks.

Section – A ($1 \times 9 = 9$ Marks)

1. Which of the following numbers is irrational?

- (a) 0.125 (b) $\sqrt{36}$
(c) $\sqrt{13}$ (d) $9/11$

2. The decimal expansion of $19/125$ is:

- (a) Terminating (b) Non-terminating recurring
(c) Non-terminating non-recurring (d) Irrational

3. The value of $(x + y)^2 - (x - y)^2$ is:

- (a) $2xy$ (b) $4xy$
(c) $x^2 - y^2$ (d) 0

4. Rationalising factor of $(4 + \sqrt{3})$ is:

- (a) $4 + \sqrt{3}$ (b) $4 - \sqrt{3}$
(c) $\sqrt{3} - 4$ (d) $1/(4 + \sqrt{3})$

True / False

5. Every natural number is a whole number. (True / False)
6. $\sqrt{7}$ is a rational number. (True / False)
7. $(a + b)^2 = a^2 + b^2$ (True / False)

Fill in the blanks

8. Decimal expansion of $1/32$ is _____.
9. $(a - b)(a + b) = \underline{\hspace{2cm}}$.

Section – B ($2 \times 3 = 6$ Marks)

10. Convert 0.5555... into p/q form.
11. Evaluate using suitable identity: 104×96
12. If $x + y = 11$ and $xy = 24$, find $x^2 + y^2$.

Section – C ($3 \times 4 = 12$ Marks)

13. Convert 0.272727... into p/q form.
14. Rationalise the denominator of $7/(\sqrt{5}+2)$.
15. If $x + 1/x = 6$, find $x^2 + 1/x^2$.
16. Expand $(a + b - c)^2$.

Section – D ($4 \times 2 = 8$ Marks)

17. Prove that $\sqrt{3}$ is irrational.
18. If $x + y + z = 0$, prove that $x^3 + y^3 + z^3 = 3xyz$.